

Minku Kim

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🌐 <https://min-ku.github.io>

Education

Oregon State University

Ph.D. Candidate in Robotics

Corvallis, OR

Current

University of Pennsylvania

M.S.E. in Mechanical Engineering and Applied Mechanics (4.0/4.0)

Philadelphia, PA

May 2025

- **Thesis:** Learning a Vision-Based Footstep Planner for Hierarchical Walking Control on Unstructured Terrain

Chung-Ang University

B.S. in Mechanical Engineering with Honors

Seoul, Korea

Feb 2023

Experience

Dynamic Robotics and Artificial intelligence Laboratory

Corvallis, OR

Graduate Research Assistant - Prof. Alan Fern

Sep 2025 – Current

- Developing loco-manipulation skills for humanoid robots using reinforcement learning and diffusion/flow-matching over contact, vision, and language inputs to enable robust, goal-conditioned whole-body interaction in complex environments

Agility

Salem, OR

Artificial Intelligence Intern

Jun 2026 – Sep 2026

- Migrated the RL training stack to MJWarp backend and extended the stack to support vision-based learning and manipulation.
- Trained and deployed vision-based manipulation policies on Digit v4 hardware using reset distributions for tote pickup, putdown, and tabletop manipulation.

Figueroa Lab, GRASP Lab

Philadelphia, PA

Graduate Research Assistant - Prof. Nadia Figueroa

Mar 2025 – Current

- Developed a real-time category-level 6D pose, shape, scale estimation, combining a flow-matching keypoint detector with Active Shape Models and a GPU-accelerated Matrix ADMM solver to achieve >100 Hz optimization for closed-loop manipulation
- Designed a real-time open-vocabulary multi-view transformer model integrating CLIP and DINOv3 for 6D pose estimation, achieving 13.7 FPS inference with an $8\times$ lower memory usage than prior baselines such as PI3

Dynamic Autonomy and Intelligent Robotics Lab, GRASP Lab

Philadelphia, PA

Graduate Research Assistant - Prof. Michael Posa

Jan 2024 – May 2025

- Designed and deployed a vision-based hierarchical controller for the *Agility Robotics Cassie* bipedal robot, integrating a high-level RL footstep planner with a low-level operational space controller
- Built a full-stack RL pipeline in *Drake* for training, sampling, and hardware deployment, and benchmarked against a vision-based MPC footstep planner, demonstrating improved velocity tracking and success rates across diverse terrains

Publications

SAGE: Semantic And Geometric Estimation of 6D Object Pose from Multi-View Observations

2026

Under review

Minku Kim[†], Ho Jin Choi[†] and Nadia Figueroa

ASM-6D: Generative Correspondences for Fast Object Pose-Shape Fitting

2026

In Equivariant Systems Workshop at RSS 2025

Under review

Ho Jin Choi[†], Minku Kim[†] and Nadia Figueroa

Humanoid Hanoi: Investigating Shared Whole-Body Control for Skill-Based Box Rearrangement

2026

In Conference on Robot Learning (CoRL) 2026

Minku Kim[†], Kuan-Chia Chen[†], Aayam Shrestha, Li Fuxin, Stefan Lee and Alan Fern

Learning a Vision-Based Footstep Planner for Hierarchical Walking Control

2025

In IEEE-RAS 24th International Conference on Humanoid Robots (Humanoids) [Oral Presentation]

Minku Kim, Brian Acosta, Pratik Chaudhari and Michael Posa

Honors and Awards

Oregon State University College of Engineering (COE) Scholarship

2025

Penn Engineering Outstanding Research Award

2025

CAU Winter Conference Da-Vinci Software Institute Excellence Award

2022

CAU Summer Conference Da-Vinci Software Institute Encouragement Award

2021

Technical Skills

Programming: Python, C/C++, MATLAB/Simulink, Git, Linux

Software/Frameworks: Pytorch, Tensorflow, OpenCV, ROS, LCM, Drake, MuJoCo, Isaac-Sim, Bazel, Docker, SLURM

Robotics: Legged Robot Control, Manipulation Control, Reinforcement Learning, Imitation Learning, Perception, Optimization